PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE C · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

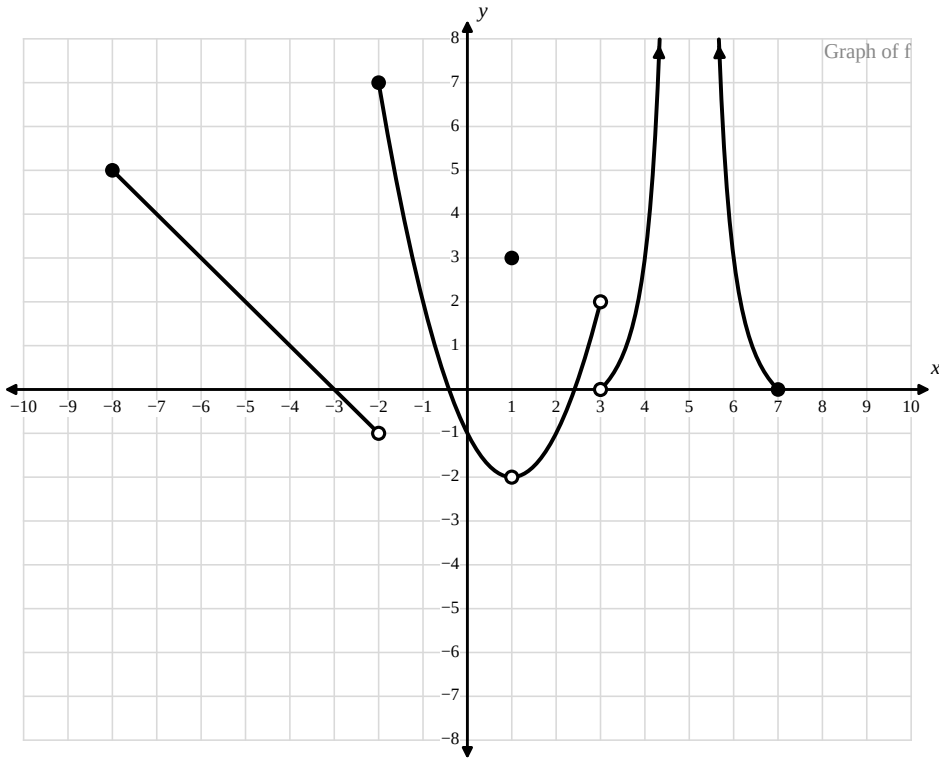
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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

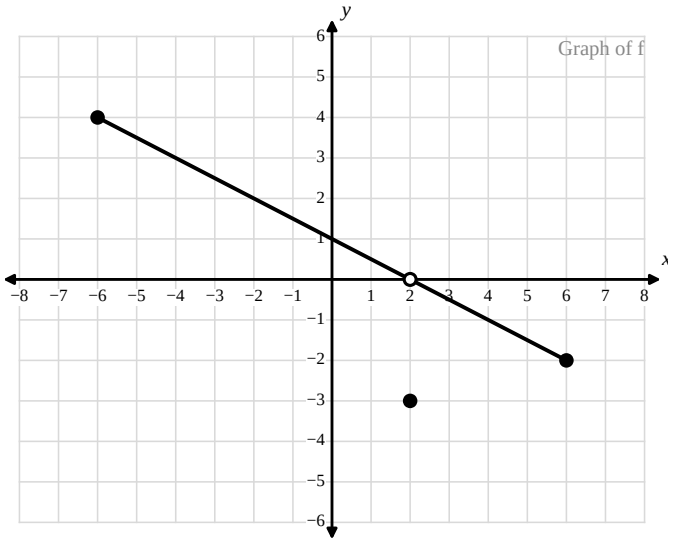


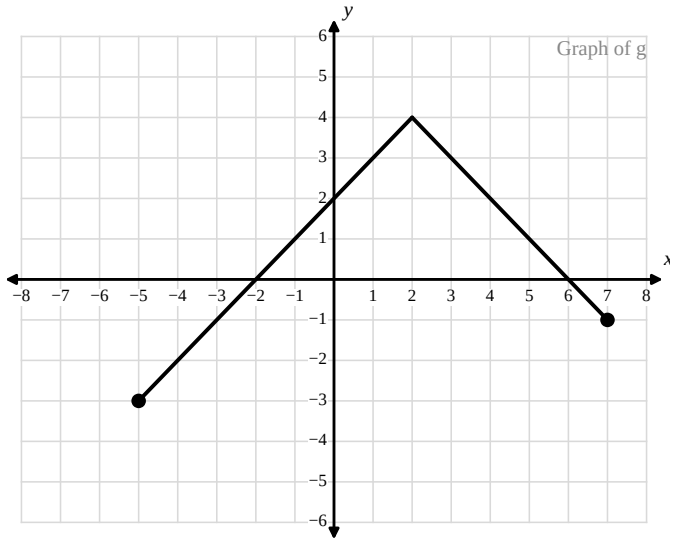
1. $\lim_{x\to 1} f(x) =$	**2.** $\lim_{x\to 3} f(x) =$	**3.** $\lim_{x\to 5} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = -1$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x \to 4} \sqrt{[g(x)]^2 - 3f(x)}$

(2 points)

6. Use the table to find each limit. If a limit does not exist, write DNE.

x	1.9	1.99	1.999	$\rightarrow 2 \leftarrow$	2.001	2.01	2.1
$h(x)$	11.41	11.9401	11.994001		6.003	6.03	6.3

$\lim_{x \to 2^+} h(x) =$ _____ $\lim_{x \to 2} h(x) =$ _____ $\lim_{x \to 2^-} h(x) =$ _____

Evaluate each limit. (2 points each)

7. $\lim_{x\to 1} f(x), \quad f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x < 1 \\ 3 - x^2, & x \geq 1 \end{cases}$

8. $\lim_{x\to \frac{1}{2}} \dfrac{2x^2 + 3x - 2}{4x^2 - 1}$

9. $\lim_{x\to 3} \dfrac{x^2 - 3x}{\sqrt{x+1} - 2}$

10. $\lim_{x\to -1} \dfrac{\frac{1}{x+2} - 1}{x^2 - 1}$

11. $\lim_{h\to 0} \dfrac{\left(-\frac{5}{2} + h\right)^2 - \left(-\frac{5}{2}\right)^2}{h}$

PCBC H Name **ANSWER KEY** **22** /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

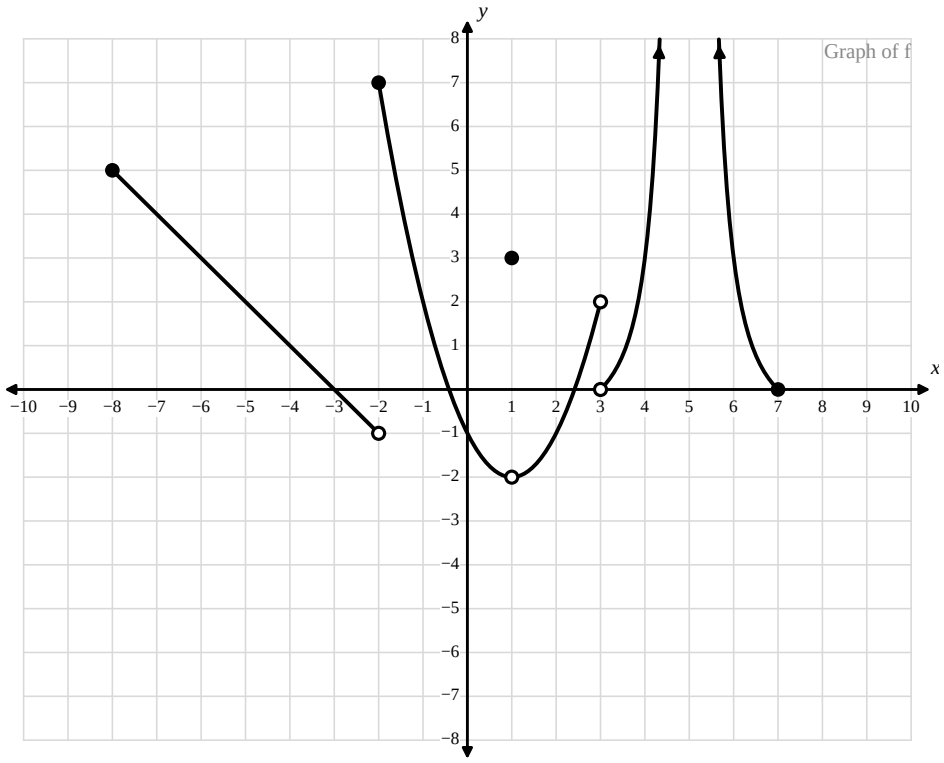
PRACTICE C · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

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SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)



1. $\lim_{x\to 1} f(x) =$

Both sides of the parabola head to the open circle at the vertex $(1, -2)$.

-2

2. $\lim_{x\to 3} f(x) =$

$\lim_{x\to 3^-} f(x) = 2$, $\lim_{x\to 3^+} f(x) = 0$

$2 \neq 0$, *so* $\lim_{x\to 3} f(x)$ *does not exist.*

DNE

3. $\lim_{x\to 5} f(x) =$

Both sides of $x = 5$ *rise without bound.*

∞

(2 points)

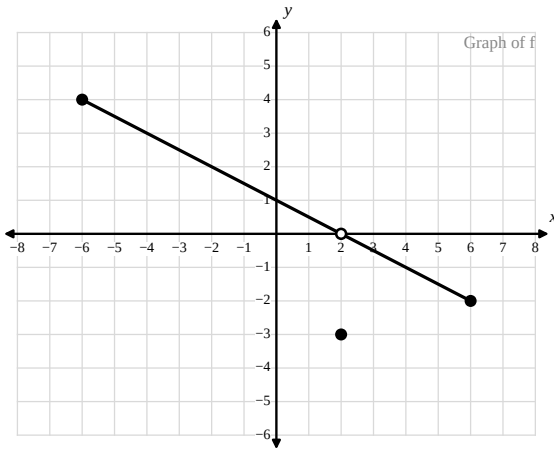
4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = -1$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

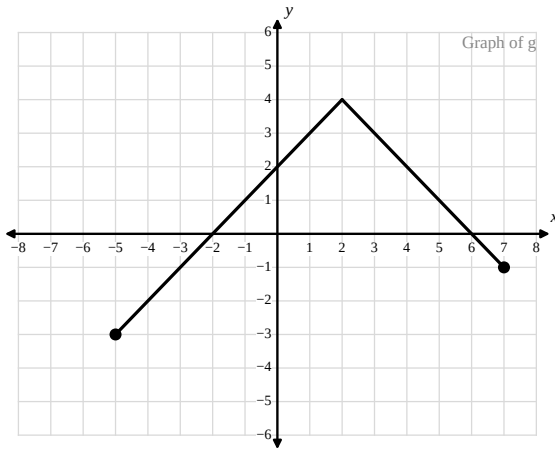
Height -1 *is reached from both sides at* $x = 0$ *and* $x = 2$.

Not $x = -2$*: only the left line heads to* -1*; the right side starts at 7.*

$a = 0,\ a = 2$

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x\to 4}\sqrt{[g(x)]^2 - 3f(x)}$

$$\lim_{x\to 4}\sqrt{[g(x)]^2 - 3f(x)} = \sqrt{\left[\lim_{x\to 4} g(x)\right]^2 - 3\lim_{x\to 4} f(x)}$$

Root (inside limit is 7 > 0), power, difference, constant multiple.

$$= \sqrt{(2)^2 - 3(-1)} = \sqrt{4+3} = \sqrt{7}$$

$\sqrt{7}$

(2 points)

6. Use the table to find each limit. If a limit does not exist, write DNE.

x	1.9	1.99	1.999	$\to 2 \leftarrow$	2.001	2.01	2.1
$h(x)$	11.41	11.9401	11.994001		6.003	6.03	6.3

Right: 6.3, 6.03, 6.003 → 6

Left: 11.41, 11.9401, 11.994001 → 12

6 ≠ 12, so the two-sided limit does not exist. The last blank asks only from the left: 12.

12

$\lim_{x\to 2^+} h(x) =$ 6 $\lim_{x\to 2} h(x) =$ DNE $\lim_{x\to 2^-} h(x) =$ 12

Evaluate each limit. (2 points each)

7. $\lim_{x\to 1} f(x), \quad f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x < 1 \\ 3 - x^2, & x \geq 1 \end{cases}$

$\lim_{x\to 1^-} \frac{x^2-1}{x-1} = \lim_{x\to 1^-} \frac{(x-1)(x+1)}{x-1} = \lim_{x\to 1^-} (x+1) = 2$

Left piece: substitution produces 0/0, so factor.

$\lim_{x\to 1^+} (3 - x^2) = 3 - 1 = 2$

$\lim_{x\to 1} f(x) = 2$

$\boxed{2}$

8. $\lim_{x\to \frac{1}{2}} \frac{2x^2 + 3x - 2}{4x^2 - 1}$

Substitution produces 0/0: indeterminate.

$\lim_{x\to \frac{1}{2}} \frac{(2x-1)(x+2)}{(2x-1)(2x+1)}$

$= \lim_{x\to \frac{1}{2}} \frac{x+2}{2x+1} = \frac{\frac{5}{2}}{2} = \frac{5}{4}$

$\boxed{\frac{5}{4}}$

9. $\lim_{x\to 3} \frac{x^2 - 3x}{\sqrt{x+1} - 2}$

Substitution produces 0/0: indeterminate.

$\lim_{x\to 3} \frac{x(x-3)\left(\sqrt{x+1}+2\right)}{(x+1)-4}$

$= \lim_{x\to 3} \frac{x(x-3)\left(\sqrt{x+1}+2\right)}{x-3}$

$= \lim_{x\to 3} x\left(\sqrt{x+1}+2\right) = 3(2+2) = 12$

$\boxed{12}$

10. $\lim_{x\to -1} \frac{\frac{1}{x+2} - 1}{x^2 - 1}$

Substitution produces 0/0: indeterminate.

LCD: x + 2

$\lim_{x\to -1} \frac{1 - (x+2)}{(x+2)(x^2-1)}$

$= \lim_{x\to -1} \frac{-(x+1)}{(x+2)(x-1)(x+1)}$

$= \lim_{x\to -1} \frac{-1}{(x+2)(x-1)} = \frac{-1}{(1)(-2)} = \frac{1}{2}$

$\boxed{\frac{1}{2}}$

11. $\lim_{h\to 0} \frac{\left(-\frac{5}{2} + h\right)^2 - \left(-\frac{5}{2}\right)^2}{h}$

Substituting h = 0 produces 0/0: indeterminate. Expand the top.

$\lim_{h\to 0} \frac{\frac{25}{4} - 5h + h^2 - \frac{25}{4}}{h}$

$= \lim_{h\to 0} \frac{h^2 - 5h}{h}$

$= \lim_{h\to 0} \frac{h(h-5)}{h}$

$= \lim_{h\to 0} (h-5) = 0 - 5 = -5$

$\boxed{-5}$